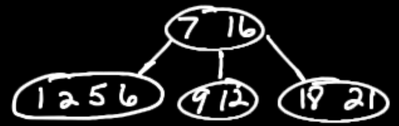
## ① B-tree

- Further generalization of a 2-3 tree
- Rudolf Bayer and Edward M. Cartwright
- 1970
- Managing index pages of large random access files
- Variants commonly used in database indexing
- uses "orders" or "degrees"
- Donald Knuth clarified the properties
  1. Every node has at most m children
  2. Every non-leaf node (except the root) has at least $\lceil m/2 \rceil$ child nodes
  3. The root has at least 2 children if it is not a leaf node
  4. A non-leaf with k children contains k-1 keys
  5. All leafs appear in the same level and carry no information

m = max # of children
= "degree" or "order"

if m=2, binary tree
if m=3, 2-3 tree
if m=4, 2-3-4 tree
k = m-1 = max # of keys




→

## ② Insert: 30 10 16 60 20 50 40 70 80 15 90 19 w/ m=6

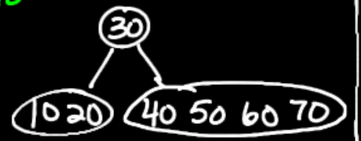
First 5 are easy



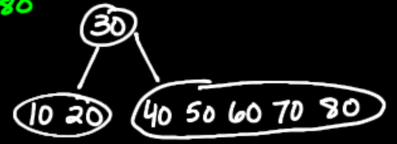

40




70



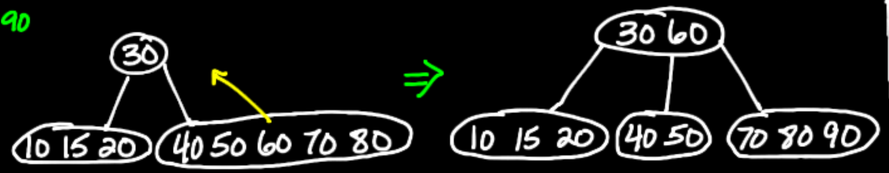

80




15




90




19

